RAG DEMO – Quick Guide

n8n + Groq + Mistral + Supabase | Travel & Expense Policy | Naji El Kotob + AI

1. What This Demo Shows

An AI chat assistant that answers questions from your own document instead of from general knowledge. It reads a Travel & Expense Policy (Markdown on GitHub), stores it in a vector database, and answers employee questions with the exact rule and the source file.

> **Why it matters:** a normal chatbot guesses policy answers. A RAG chatbot looks up your document first, so answers are specific, current and checkable – and it says "I don't know" when the document has no answer.

2. The Workflow at a Glance

The workflow has two separate parts that share one database:

Part A – Load the document (run once, or after each edit)

Node	What it does
Execute workflow	Manual start button.
Config	Holds the raw GitHub URL of the Markdown file. The only value you edit.
Delete old chunks	Removes this file's old rows from Supabase so re-running never creates duplicates.
Get file from GitHub	Downloads the Markdown text (HTTP Request, no login needed for a public repo).
Recursive Text Splitter	Cuts the text into ~1,000-character pieces with 200 characters of overlap, so no rule is cut in half.
Default Data Loader	Wraps each piece and tags it with file_url and file_name (used for citations).
Embeddings Mistral	Turns each piece into a list of 1,024 numbers that capture its meaning.
Supabase Vector Store (insert)	Saves the text + numbers into the documents table.

Part B – Chat (runs on every question)

Node	What it does
When chat message received	Opens the n8n chat window and receives the question.
AI Agent	The decision-maker. Its system prompt forces it to search first and answer only from the results.
Groq Chat Model	The language model (llama-3.3-70b-versatile) that reads the results and writes the answer.
Simple Memory	Remembers the last 10 messages, so follow-up questions work.
Supabase Vector Store (search)	The knowledge_base tool: finds the 5 most relevant pieces for the question.
Embeddings Mistral (shared)	Converts the question into numbers using the same model as Part A.

> **Key rule:** Part A and Part B must use the same embedding model. That is why one Mistral node feeds both, and why the table size (1,024) matches Mistral. Change the model and you must rebuild the table and reload the document.

3. Setup Checklist

1. **Supabase table:** SQL Editor → New query → paste rag-supabase-setup.sql → Run. Confirm the documents table appears in Table Editor.
2. **GitHub file:** create a public repo, upload travel-expense-policy.md, open it, click Raw and copy the URL.
3. **Import workflow:** in n8n, Workflows → Import from File → rag-free-demo.json.
4. **Credentials (4):** Groq API key (console.groq.com), Mistral API key (console.mistral.ai, Experiment plan), Supabase API (project URL + service_role key), Postgres (Supabase connection details – use the Session pooler host if the direct host fails).
5. **Config node:** paste the raw GitHub URL into fileUrl.
6. **Load:** click Execute workflow. In Supabase Table Editor you should now see rows in documents.
7. **Chat:** click Open chat in n8n and ask the test questions below.

4. Test Questions

Use these in order during the demo. The "Shows" column tells the audience what each question proves.

#	Question	Expected answer	Shows
1	What class can I fly on a 7-hour flight?	Premium Economy	Basic lookup
2	What is the hotel limit in a Tier A city?	USD 280 per night incl. taxes	Table lookup
3	What is the per diem for a travel day in a Tier C city?	75% of USD 45 = USD 33.75	Combines two facts
4	Breakfast is included at my hotel. How much is deducted?	15% of the daily rate	Detail rule
5	Who approves a USD 8,000 trip?	Department Head + Finance Director	Threshold logic
6	Can I claim a speeding ticket from a rental car?	No, never reimbursed	Negative answer
7	I submitted my claim 45 days late. What happens?	Needs Department Head approval (31–90 days)	Range rule
8	Do I need a receipt for a USD 20 taxi?	No – under USD 25, max USD 75 per trip	Edge case
9	I stayed with family. Can I claim anything?	Yes, USD 30 per night, no receipt	Less obvious rule
10	What is the mileage rate for my own car?	USD 0.45 per km	Simple number
11	And can I claim parking on top of that?	Yes, with receipts	Memory (follow-up)

#	Question	Expected answer	Shows
12	Can I keep my airline miles?	Yes, but they must not drive the airline choice	Nuance
13	What is the parental leave policy?	"I couldn't find that in the documents."	No guessing

> **Demo tip:** ask question 13 last. Seeing the assistant refuse to invent an answer is the moment that convinces business audiences.

5. Updating the Knowledge

- Edit the Markdown on GitHub (for example, change the mileage rate), then click Execute workflow again.
- The Delete old chunks step removes the old version first, so the assistant always answers from the latest text.
- To add another document, change fileUrl and run again – each file keeps its own rows.

6. Troubleshooting

Problem	Likely cause and fix
Insert fails: "different vector dimensions"	Table size does not match the embedding model. Re-run supabase-setup.sql (1,024 for Mistral).
"429 Too many requests"	Free-tier rate limit (usually Mistral). Wait a few seconds between questions.
Agent answers without searching	Check the knowledge_base tool is connected to the Agent's Tool slot and the system prompt is in place.
Answer says "couldn't find" for a known rule	Document not loaded, or loaded from the wrong URL. Check rows exist in Supabase.
GitHub node returns 404	Repo is private, or the URL is the normal page instead of the Raw link.
Postgres credential will not connect	Use the Session pooler host and port from Supabase → Connect, and enable SSL.
Duplicate answers / repeated chunks	Old rows from before the delete step. Run: delete from documents; then reload.

Free-tier limits change over time – check each provider's console before a live session.